Grounded Citation Ledger

esperanto-research — a citation-verified review of OPUS23 | ESPERANTO

Generated 2026-10-30 from `citations/ledger.json` (grounded-citations workflow)

Method

Every citation in this project's reports resolves to an entry in the mechanical ledger (file `citations/ledger.json`); ids are issued only at retrieval, never from memory. This document renders the full ledger: **51 registered sources**, of which the [1, 2, 4, 5] platform sources carry **16 verbatim evidence quotes** — each machine-checked ("`sources.py quote --from`") to appear verbatim, whitespace- and markup-tolerant, in the fetched page text preserved under `extracts/`. All pages were accessed 2026-10-30. Quotation text below is reproduced exactly as attached (raw page markup such as `**` may be retained inside quotes). Regenerate with `citations/build_ledger_pdf.py`.

Platform sources — evidence-carrying

[1] OPUS23 | ESPERANTO · Datapunk Bioinformatics (platform landing page)
https://www.datapunk.net *(accessed 2026-10-30)*
Verbatim evidence: — *"A 22,800-taxon knowledge base behind every stool report."* — *"390 explained markers, 3,200+ metabolites"* — *"Drafted by a dedicated large language model in the background; every claim carries a citation you can click."* — *"Opie is our literature agent."* — *"Pay as you go, with tokens."* — *"It does not diagnose, treat or prescribe"* — *"ABO, secretor (FUT2) and Lewis status"* — *"389** multi-SNP algorithms"* — *"4,500+** curated SNPs"*

[2] Datapunk Opus23 Pro (legacy splash)
https://www.datapunk.net/opus23 *(accessed 2026-10-30)*
Verbatim evidence: — *"Over 200 clinicians worldwide use Opus23"* — *"Each license is for one client"* — *"report over 300 potential adverse reactions"*

[3] Datapunk Open Source Apps
https://datapunk.net/tlfd *(accessed 2026-10-30)*

[4] Opus23 Pro Help Docs — Welcome
https://www.datapunk.net/opus23/docs/index.pl?WELCOME *(accessed 2026-10-30)*
Verbatim evidence: — *"runs over 300 algorithms on multi-snp clusters"* — *"exceeds HIPAA standards"* — *"patients at the University of Bridgeport, CT"*

[5] Opus23 Developer Blog — About Opus 23
https://datapunk.net/opus23blog/about_opus23 *(accessed 2026-10-30)*
Verbatim evidence: — *"Center of Excellence in Generative Medicine (COEGM) to analyze and interpret raw genomic data"*

Aggregated dossier sources — competitor verification (03)

[6] Jona Clinical Gut Microbiome Analysis Advanced
https://jona.health/products/jona-clinical-gut-microbiome-analysis-advanced *(accessed 2026-10-30)*

[7] DNA Testing & Precision Medicine Platform for Medical Doctors
https://promicsedge.com/medical-doctors *(accessed 2026-10-30)*

[8] Lite - OmicsEdge
https://promicsedge.com/lite *(accessed 2026-10-30)*

[9] Store - Tiny Health
https://www.tinyhealth.com/store *(accessed 2026-10-30)*

[10] Core Panels | Fagron Genomics US
https://gxsciences.com/nutrigenomics *(accessed 2026-10-30)*

[11] Pro7 Panel | Fagron Genomics US
https://gxsciences.com/nutrigenomics/pro7-panel *(accessed 2026-10-30)*

[12] Best AI Scribe for Functional Medicine (2026) | DeepCura Resources
https://www.deepcura.com/resources/best-ai-scribe-for-functional-medicine *(accessed 2026-10-30)*

[13] DeepCura — Beyond the EHR: The Electronic Health Workforce
https://www.deepcura.com *(accessed 2026-10-30)*

[14] Microbiome Analysis Software - Metabolon
https://www.metabolon.com/bioinformatics-software/microbiome-software *(accessed 2026-10-30)*

[15] 3MCor: An Integrative Web Server for Metabolome-Microbiome-Metadata Correlation Analysis (preprint)
https://pdfs.semanticscholar.org/e7bc/6cd38db249ae0f9448dc29195d3c8ef52cb5.pdf *(accessed 2026-10-30)*

[16] AmiAge Clock
https://3mcor.cn *(accessed 2026-10-30)*

[18] MMIP — Microbiome Metabolome Integration Platform
https://bioinfo.jisiasr.org/mmip/index.html *(accessed 2026-10-30)*

[19] FDX Method - Functional DX
https://functionaldx.com/fdx-method *(accessed 2026-10-30)*

[20] Home Functional DX
https://functionaldx.com *(accessed 2026-10-30)*

Aggregated dossier sources — science-flags literature check (05)

[17] Microbiome Metabolome Integration Platform (MMIP) — PubMed PMID 37771003
https://pubmed.ncbi.nlm.nih.gov/37771003 *(accessed 2026-10-30)*

[21] Identification of human zonulin, a physiological modulator of tight junctions, as prehaptoglobin-2 (PNAS 2009)
https://pubmed.ncbi.nlm.nih.gov/19805376 *(accessed 2026-10-30)*

[22] Zonulin, regulation of tight junctions, and autoimmune diseases (Ann NY Acad Sci 2012)
https://pubmed.ncbi.nlm.nih.gov/22731712 *(accessed 2026-10-30)*

[23] Celiac disease (Curr Opin Gastroenterol 2009)
https://pubmed.ncbi.nlm.nih.gov/19122520 *(accessed 2026-10-30)*

[24] A purified membrane protein from Akkermansia muciniphila or the pasteurized bacterium improves metabolism in obese and diabetic mice (Nat Med 2017)
https://pubmed.ncbi.nlm.nih.gov/27892954 *(accessed 2026-10-30)*

[25] Commensal Akkermansia muciniphila exacerbates gut inflammation in Salmonella Typhimurium-infected gnotobiotic mice (PLoS ONE 2013)
https://pubmed.ncbi.nlm.nih.gov/24040367 *(accessed 2026-10-30)*

[26] Faecalibacterium prausnitzii is an anti-inflammatory commensal bacterium identified by gut microbiota analysis of Crohn disease patients (PNAS 2008)
https://pubmed.ncbi.nlm.nih.gov/18936492 *(accessed 2026-10-30)*

[27] Bifidobacteria and Butyrate-Producing Colon Bacteria: Importance and Strategies for Their Stimulation in the Human Gut (Front Microbiol 2016)
https://pubmed.ncbi.nlm.nih.gov/27446020 *(accessed 2026-10-30)*

[28] Widely Used Commercial ELISA Does Not Detect Precursor of Haptoglobin2, but Recognizes Properdin as a Potential Second Member of the Zonulin Family (Front Endocrinol 2018)
https://pubmed.ncbi.nlm.nih.gov/29459849 *(accessed 2026-10-30)*

[29] A fast, cheap, and easy protocol for celiac disease HLA haplotype typing using buccal swabs (Methods Cell Biol 2023)
https://pubmed.ncbi.nlm.nih.gov/37625876 *(accessed 2026-10-30)*

[30] Effective detection of human leukocyte antigen risk alleles in celiac disease using tag single nucleotide polymorphisms (PLoS ONE 2008)
https://pubmed.ncbi.nlm.nih.gov/18509540 *(accessed 2026-10-30)*

[31] Cost-effective HLA typing with tagging SNPs predicts celiac disease risk haplotypes in the Finnish, Hungarian, and Italian populations (Immunogenetics 2009)
https://pubmed.ncbi.nlm.nih.gov/19255754 *(accessed 2026-10-30)*

[32] ESPGHAN guidelines for the diagnosis of coeliac disease (J Pediatr Gastroenterol Nutr 2012)
https://pubmed.ncbi.nlm.nih.gov/22197856 *(accessed 2026-10-30)*

[33] ESPGHAN Guidelines for Diagnosing Coeliac Disease 2020 (J Pediatr Gastroenterol Nutr)
https://pubmed.ncbi.nlm.nih.gov/31568151 *(accessed 2026-10-30)*

[34] The GenoChip: a new tool for genetic anthropology (Genome Biol Evol 2013)
https://pubmed.ncbi.nlm.nih.gov/23666864 *(accessed 2026-10-30)*

[35] Mitochondrial DNA Haplogroup K Is Protective Against Autism Spectrum Disorder Risk (JAACAP 2024)
https://pubmed.ncbi.nlm.nih.gov/38072244 *(accessed 2026-10-30)*

[36] Examining the role of common and rare mitochondrial variants in schizophrenia (PLoS ONE

2018)
https://pubmed.ncbi.nlm.nih.gov/29370225 *(accessed 2026-10-30)*

[37] The association between common vitamin D receptor gene variations and osteoporosis: a participant-level meta-analysis (Ann Intern Med 2006)
https://pubmed.ncbi.nlm.nih.gov/16908916 *(accessed 2026-10-30)*

[38] BsmI, TaqI, ApaI and FokI polymorphisms in the vitamin D receptor (VDR) gene and risk of fracture in Caucasians: a meta-analysis (Bone 2011)
https://pubmed.ncbi.nlm.nih.gov/20601302 *(accessed 2026-10-30)*

[39] Do Vitamin D receptor gene polymorphisms affect bone mass density in men? (Ageing Res Rev 2022)
https://pubmed.ncbi.nlm.nih.gov/35063697 *(accessed 2026-10-30)*

[40] Common genetic determinants of vitamin D insufficiency: a genome-wide association study (Lancet 2010)
https://pubmed.ncbi.nlm.nih.gov/20541252 *(accessed 2026-10-30)*

[41] A second common mutation in the methylenetetrahydrofolate reductase gene: an additional risk factor for neural-tube defects? (Am J Hum Genet 1998)
https://pubmed.ncbi.nlm.nih.gov/9545395 *(accessed 2026-10-30)*

[42] MTHFR 677C–>T polymorphism and risk of coronary heart disease: a meta-analysis (JAMA 2002)
https://pubmed.ncbi.nlm.nih.gov/12387655 *(accessed 2026-10-30)*

[43] ACMG Practice Guideline: lack of evidence for MTHFR polymorphism testing (Genet Med 2013)
https://pubmed.ncbi.nlm.nih.gov/23288205 *(accessed 2026-10-30)*

[44] Proposed guidelines to evaluate scientific validity and evidence for genotype-based dietary advice (Genes Nutr 2017)
https://pubmed.ncbi.nlm.nih.gov/29270237 *(accessed 2026-10-30)*

[45] APOA2, dietary fat, and body mass index: replication of a gene-diet interaction in 3 independent populations (Arch Intern Med 2009)
https://pubmed.ncbi.nlm.nih.gov/19901143 *(accessed 2026-10-30)*

[46] Apolipoprotein A2 polymorphism interacts with intakes of dairy foods to influence body weight in 2 U.S. populations
https://pubmed.ncbi.nlm.nih.gov/24108135 *(accessed 2026-10-30)*

[47] Adiposity-related evidence for APOA2, FTO, and MC4R: a PRISMA-ScR scoping review of recent human studies (2019-2025)
https://pubmed.ncbi.nlm.nih.gov/42950483 *(accessed 2026-10-30)*

[48] Genes involved in the regulation of intestinal permeability and their role in ulcerative colitis (J Dig Dis 2016)
https://pubmed.ncbi.nlm.nih.gov/26512799 *(accessed 2026-10-30)*

[49] Polymorphisms in E-cadherin (CDH1) result in a mis-localised cytoplasmic protein that is associated with Crohn's disease (Gut 2009)
https://pubmed.ncbi.nlm.nih.gov/19398441 *(accessed 2026-10-30)*

[**50**] Polygenic Risk Scores for Cardiovascular Disease: A Scientific Statement From the American Heart Association (Circulation 2022)
https://pubmed.ncbi.nlm.nih.gov/35862132 *(accessed 2026-10-30)*

[**51**] Clinical use of current polygenic risk scores may exacerbate health disparities (Nat Genet 2019)
https://pubmed.ncbi.nlm.nih.gov/30926966 *(accessed 2026-10-30)*